Cohomology of products and blow-ups of projective space

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Throughout this note all varieties are considered over the complex numbers.

We have shown that:

$$H^*(\mathrm{Bl}_p \mathbb{P}^2) = \mathbb{Z}[h,e]/(he, h^3, h^2+e^2), \qquad |h| = |e| = 2, \tag{1}$$
$$H^*(\mathbb{P}^1 \times \mathbb{P}^1) = \mathbb{Z}[\alpha,\beta]/(\alpha^2, \beta^2), \qquad |\alpha| = |\beta| = 2 \tag{2}$$

The universal coefficient theorem states that for any space X and any (commutative) ring R, there is a short exact sequence,

$$0 \to \mathrm{Ext}(H_{n-1}(X), R) \to H^n(X;R) \to \mathrm{Hom}(H_n(X), R) \to 0. \tag{3}$$

Since both $H_*(\mathrm{Bl}_p \mathbb{P}^2)$ and $H_*(\mathbb{P}^1 \times \mathbb{P}^1)$ are finite rank free $\mathbb{Z}$-modules,

$$\mathrm{Ext}(H_{n-1}(X), R) = 0, \quad \mathrm{Hom}(H_n(X), R) \cong R \otimes H_n(X) \tag{4}$$

Therefore there are isomorphisms $H^n(X;R) \to R \otimes H_n(X)$, natural in R. These induce an isomorphism $R \otimes H^*(X) \to H^*(X;R)$ for $X = \mathrm{Bl}_p \mathbb{P}^2$ or $\mathbb{P}^1 \times \mathbb{P}^1$, compatible with the cup product. I feel like you probably believe me already or don't care, so I'm not going to explain the word "induce" in the last sentence.

Theorem 1. *If R is a commutative ring then there is an isomoprhism of graded rings from $H^*(\mathrm{Bl}_p \mathbb{P}^2; R)$ to $H^*(\mathbb{P}^1 \times \mathbb{P}^1; R)$ if and only if $2 \in R^\times$.*

Proof. Let ϕ be an isomorphism of graded rings $H^*(\mathrm{Bl}_p \mathbb{P}^2; R) \to H^*(\mathbb{P}^1 \times \mathbb{P}^1; R)$. Since $H^4(\mathrm{Bl}_p \mathbb{P}^2; R) = R \cdot h^2$ and $H^4(\mathbb{P}^1 \times \mathbb{P}^1; R) = R \cdot \alpha\beta$, $\phi(h^2) = u\alpha\beta$ for some $u \in R^\times$. Now, any element $x \in H^2(\mathbb{P}^1 \times \mathbb{P}^1; R)$ can be written $x = m\alpha + n\beta$ for some $m, n \in R$, and $(m\alpha + n\beta)^2 = 2mn\alpha\beta$. But then $\phi(h)^2 = 2k\alpha\beta$ for some $k \in R$, which implies that $2k = u \in R^\times$, and this implies $2 \in R^\times$.

Now suppose that $2 \in R^\times$. We define the map ϕ on the basis $1, h, e, h^2$ by

$$(1, h, e, h^2) \mapsto (1, \alpha+\beta, \alpha-\beta, 2\alpha\beta). \tag{5}$$

It is clear that since $2 \in R^\times$, ϕ is a graded linear isomorphism, so the only question is whether it is a ring homomorphism. Zeroethly, $\phi(1) = 1$. Secondly,

$$\phi(h)\phi(e) = (\alpha + \beta)(\alpha - \beta) = \alpha^2 - \beta^2 = 0 = \phi(he) \tag{6}$$

$$\phi(h)^2 = (\alpha + \beta)^2 = 2\alpha\beta = \phi(h^2) \tag{7}$$

$$\phi(e)^2 = (\alpha - \beta)^2 = -2\alpha\beta = \phi(-h^2) = \phi(e^2) \tag{8}$$

$$\phi(h)^3 = (\alpha + \beta)^3 = \alpha^3 + 3\alpha^2\beta + 3\alpha\beta^2 + \beta^3 = 0 = \phi(h^3). \tag{9}$$

Therefore ϕ is a ring homomorphism, so the theorem is proved.

□